

CAPSTONE PROJECT – ENTERTAINMENT PREFERENCE ANALYSIS

Global Music Streaming Listening Preference

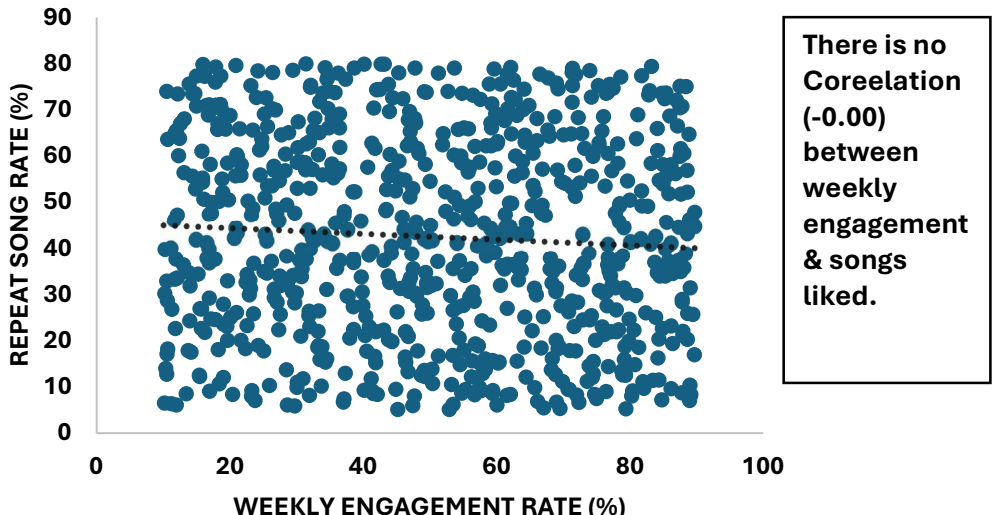
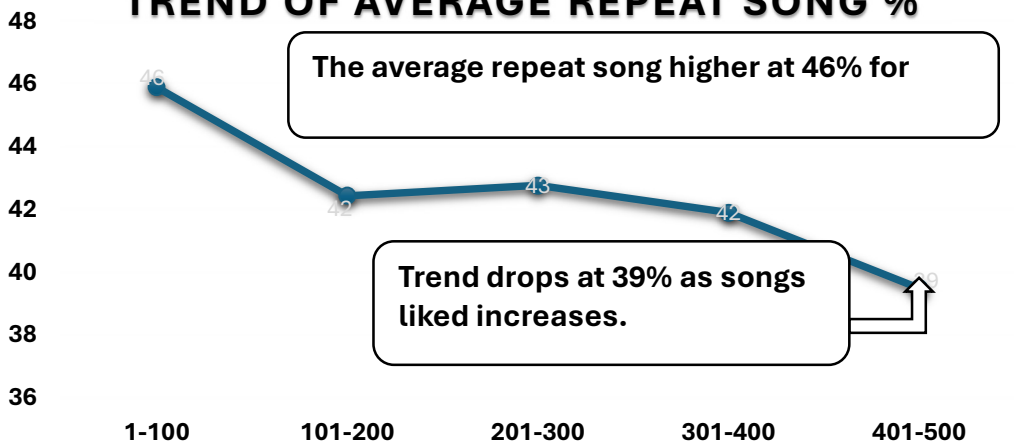
1. Project Overview

This project analysis global music streaming listener preferences & engagement using Microsoft excel. The analysis will examine listener demographics, streaming platforms, genre, most played artist, listening behaviour, subscription types, weekly engagement metrics to identify meaningful patterns and business insights.

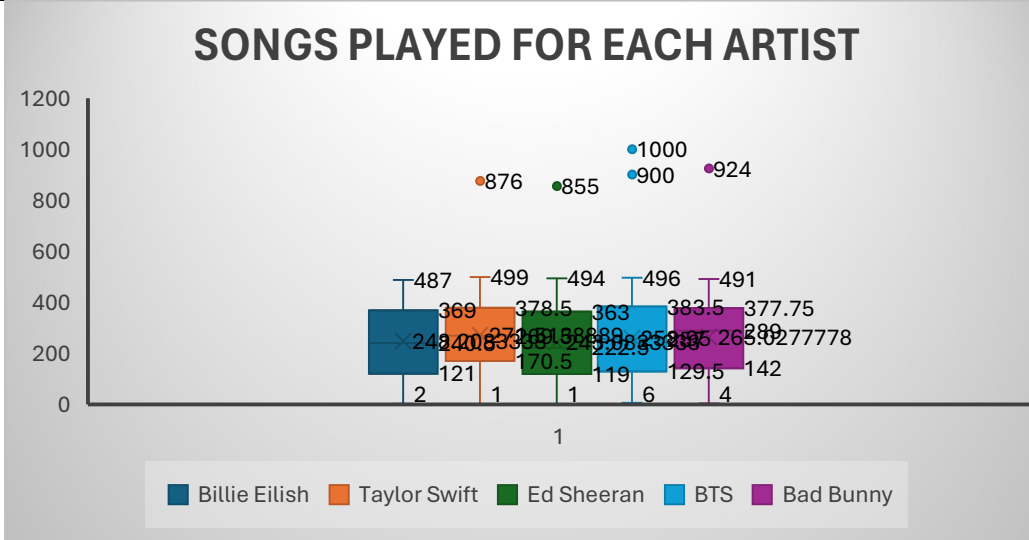
2. Business Objective

- ***Identify the most preferred streaming platforms, genre, artists & listening periods.***
- ***Measure listener engagement & compare behaviour across subscription types & age groups.***
- ***Identify relationship between listening time, songs liked, discover weekly engagement & repeat songs rate***

BUSINESS QUESTIONS	CHARTS	INTERPRETATION																								
What percentage of listeners prefers each artist ?	'LISTENERS PREFERENCE BY ARTIST A pie chart titled 'LISTENERS PREFERENCE BY ARTIST' showing the percentage of listeners who prefer each of five artists. The segments are: Bad Bunny (22%, grey), Taylor Swift (21%, purple), Ed Sheeran (20%, light blue), BTS (18%, green), and Billie Eilish (19%, orange). A legend on the right lists the artists with their corresponding colors. An arrow points from the BTS segment to its label. <table><tr><th>Artist</th><th>Percentage</th></tr><tr><td>Bad Bunny</td><td>22%</td></tr><tr><td>Taylor Swift</td><td>21%</td></tr><tr><td>Ed Sheeran</td><td>20%</td></tr><tr><td>BTS</td><td>18%</td></tr><tr><td>Billie Eilish</td><td>19%</td></tr></table>	Artist	Percentage	Bad Bunny	22%	Taylor Swift	21%	Ed Sheeran	20%	BTS	18%	Billie Eilish	19%	'Listeners favorite artist is "BAD BUNNY". "BTS is less preferred by listeners.												
Artist	Percentage																									
Bad Bunny	22%																									
Taylor Swift	21%																									
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Billie Eilish	19%																									
How does listening time affects repeat songs rate for each artist?	REPEAT SONG RATE BY ARTIST & LISTENING TIME A bar chart titled 'REPEAT SONG RATE BY ARTIST & LISTENING TIME' showing the repeat song rate for five artists across three listening times: Afternoon (blue), Morning (orange), and Night (green). The y-axis represents the repeat rate from 36 to 48. The x-axis lists the artists. A text box on the right states: 'Morning has highest repeat songs rate for each artists, while night'. <table><tr><th>Artist</th><th>Afternoon</th><th>Morning</th><th>Night</th></tr><tr><td>Bad Bunny</td><td>45</td><td>46</td><td>41</td></tr><tr><td>Billie Eilish</td><td>42</td><td>45</td><td>45</td></tr><tr><td>BTS</td><td>41</td><td>44</td><td>42</td></tr><tr><td>Ed Sheeran</td><td>43</td><td>41</td><td>40</td></tr><tr><td>Taylor Swift</td><td>40</td><td>41</td><td>41</td></tr></table>	Artist	Afternoon	Morning	Night	Bad Bunny	45	46	41	Billie Eilish	42	45	45	BTS	41	44	42	Ed Sheeran	43	41	40	Taylor Swift	40	41	41	Morning has highest repeat songs rate for each artists, while night shows lowest repeat rate
Artist	Afternoon	Morning	Night																							
Bad Bunny	45	46	41																							
Billie Eilish	42	45	45																							
BTS	41	44	42																							
Ed Sheeran	43	41	40																							
Taylor Swift	40	41	41																							

<i>Does discover weekly engagement relate to repeat song rate?</i>	WEEKLY ENGAGEMENT Vs. REPEAT SONG RATE  There is no Coreelation (-0.00) between weekly engagement & songs liked.	<i>There is no Coreelation (-0.00) between weekly engagement & songs liked.</i>												
<i>What is the trend in repeat song rate relative to the no.of songs liked?</i>	TREND OF AVERAGE REPEAT SONG %  The average repeat song higher at 46% for Trend drops at 39% as songs liked increases. <table><tr><th>Range of Songs Liked</th><th>Average Repeat Song %</th></tr><tr><td>1-100</td><td>46</td></tr><tr><td>101-200</td><td>42</td></tr><tr><td>201-300</td><td>43</td></tr><tr><td>301-400</td><td>42</td></tr><tr><td>401-500</td><td>39</td></tr></table>	Range of Songs Liked	Average Repeat Song %	1-100	46	101-200	42	201-300	43	301-400	42	401-500	39	<i>The average repeat song higher at 46% for users with few liked songs (1-100).</i>
Range of Songs Liked	Average Repeat Song %													
1-100	46													
101-200	42													
201-300	43													
301-400	42													
401-500	39													

Calculate mean, median, mode for no. of songs played for each artist?



Except billie eilish, others have extreme upper outliers. Overall data is right skewed

4.Data Cleaning Steps

- Import Dataset into excel
- Identify and handle missing values
- **Remove duplicates**
- **Validate the data**
- **Handle outliers**
- **Prepare final dataset**

5.Project Methodology

- ❖ **Explore & Clean (apply excel functions & clean dataset)**
- ❖ **Analyze (use pivot tables, statistics, outlier analysis, spread & correlation)**

❖ ***Visualize & Tell the story (create charts, identify key insights & build visual dashboard)***

6.Deliverables

The final deliverables will include:

- 1. Cleaned/preprocessed dataset***
- 2. Data analysis file and brief report***
- 3. Visual Storytelling Dashboard***
- 4. Summary***
- 5.Conclusion***

7.Roles, Responsibilities & communication:

- All Members: Review the work together, discuss findings, make required changes & ensure final project is accurate & complete.***
- Communication: The team will communicate through the group chat/ whatsapp & discuss progress & tasks regularly.***

Preferred Artist

LISTENER REPEAT RATES & ARTIST PERFORMANCE

AVG MIN STREAMED PER DAY

309.9

TOTAL NO. OF SONGS LIKED

193255

WEEKLY ENGAGEMENT RATE(%)

50.55

REPEAT SONGS RATE (%)

42.444

CORREL

-0.064

STREAMING INSIGHTS:-

Tracking global music streaming trends to understand listener engagement, repeat song trend, track performance & peak listening slots.

Most Played Artist

Bad Bunny

Billie Eilish

BTS

Ed Sheeran

Taylor Swift

Repeat Song Ra...

5.11

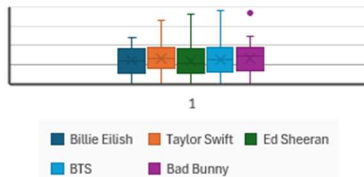
5.16

5.25

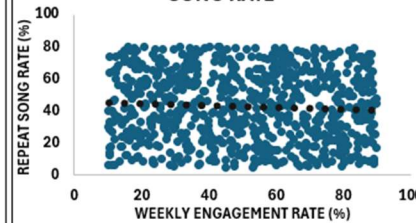
5.34

5.51

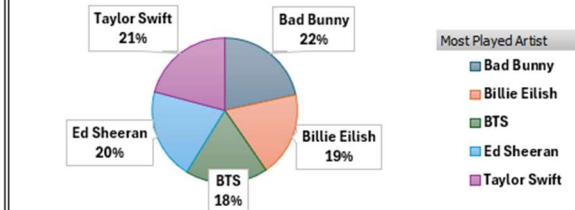
SONGS PLAYED FOR EACH ARTIST



WEEKLY ENGAGEMENT Vs. REPEAT SONG RATE

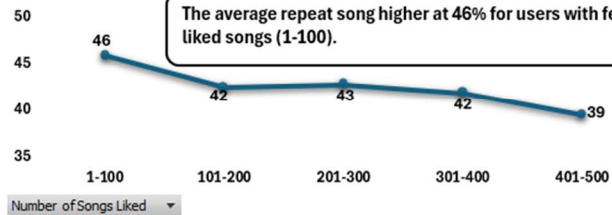


'LISTENERS' PREFERENCE BY ARTIST



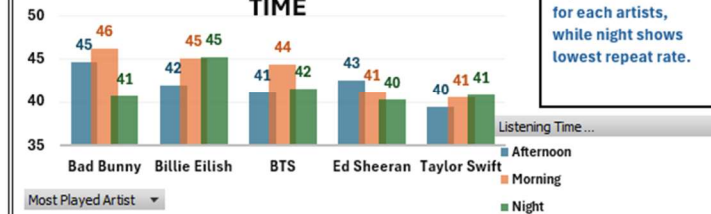
Average of Repeat Song Rate (%)

TREND OF AVERAGE REPEAT SONG %



Average of Repeat Song Rate (%)

REPEAT SONG RATE BY ARTIST & LISTENING TIME



Streaming platform

Most preferred streaming platform for the entertainment

To identify user preferences, subscription pattern and engagement behavior cross the streaming pattern

Subscription Ty...

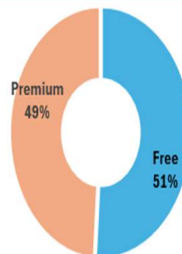
Free

Premium

Streaming Platf...

Amazon Music

Apple Music

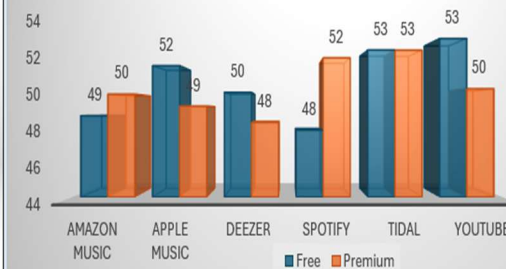


Most popular streaming platform

Most preferred platform is Tidal and least is amazon music

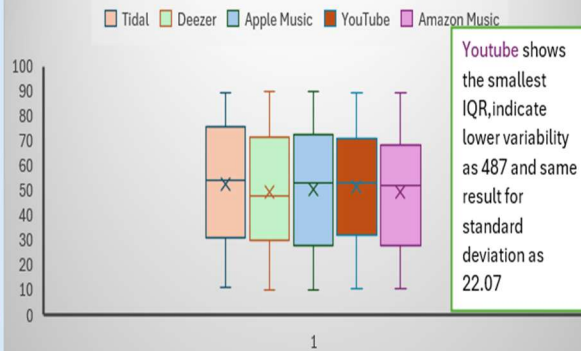


Streaming platform- free vs premium

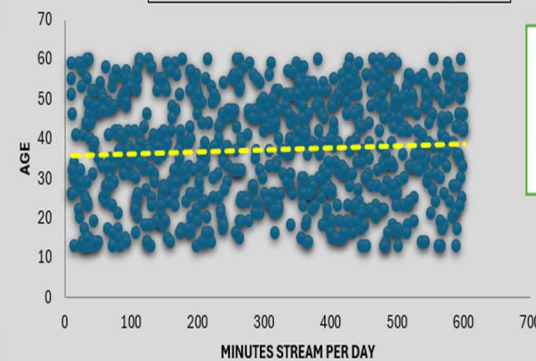


Equal distribution of subscription type
Tidal is the most popular platform for both the user.
Youtube is the more preferred by free users

Most consistent streaming platform



Minutes stream per day by age



310

Avg minutes stream per day

YouTube

Most Consistant platform

51%

Free Users

49%

Premium Users

No Correlation

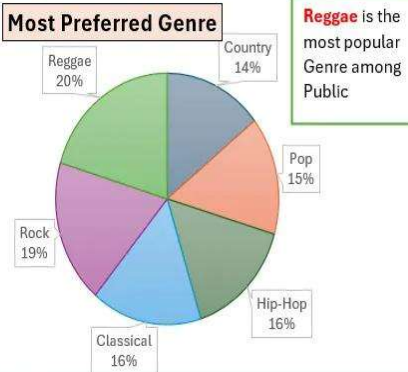
Correl:0.05
Between Age group and stream per day

Major Music Genres

ENTRTAINMENT PREFERENCE - GENRE ANALYSIS

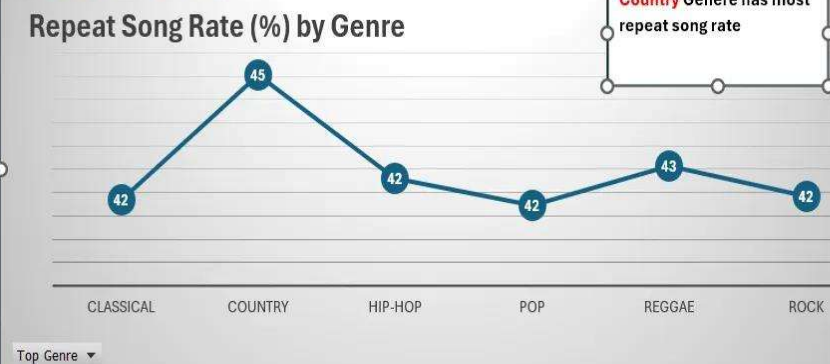
Count of Top Genre

Most Preferred Genre



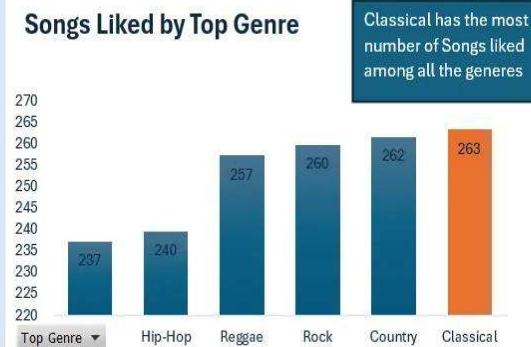
Average of Repeat Song Rate (%)

Repeat Song Rate (%) by Genre

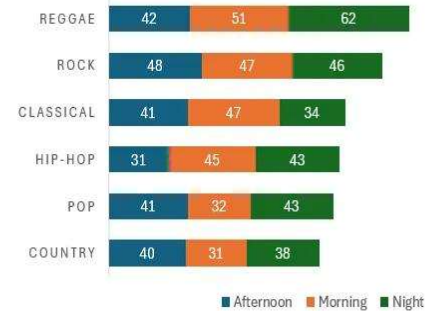


Average of Number of Songs Liked

Songs Liked by Top Genre



GENRE PREFERENCE BY TIME



Average Repeat Song Rate

42

Total No. of Songs Liked

194157

Weekly Engagement

51

Average of Age

37

Average Minutes Streamed per Day

309